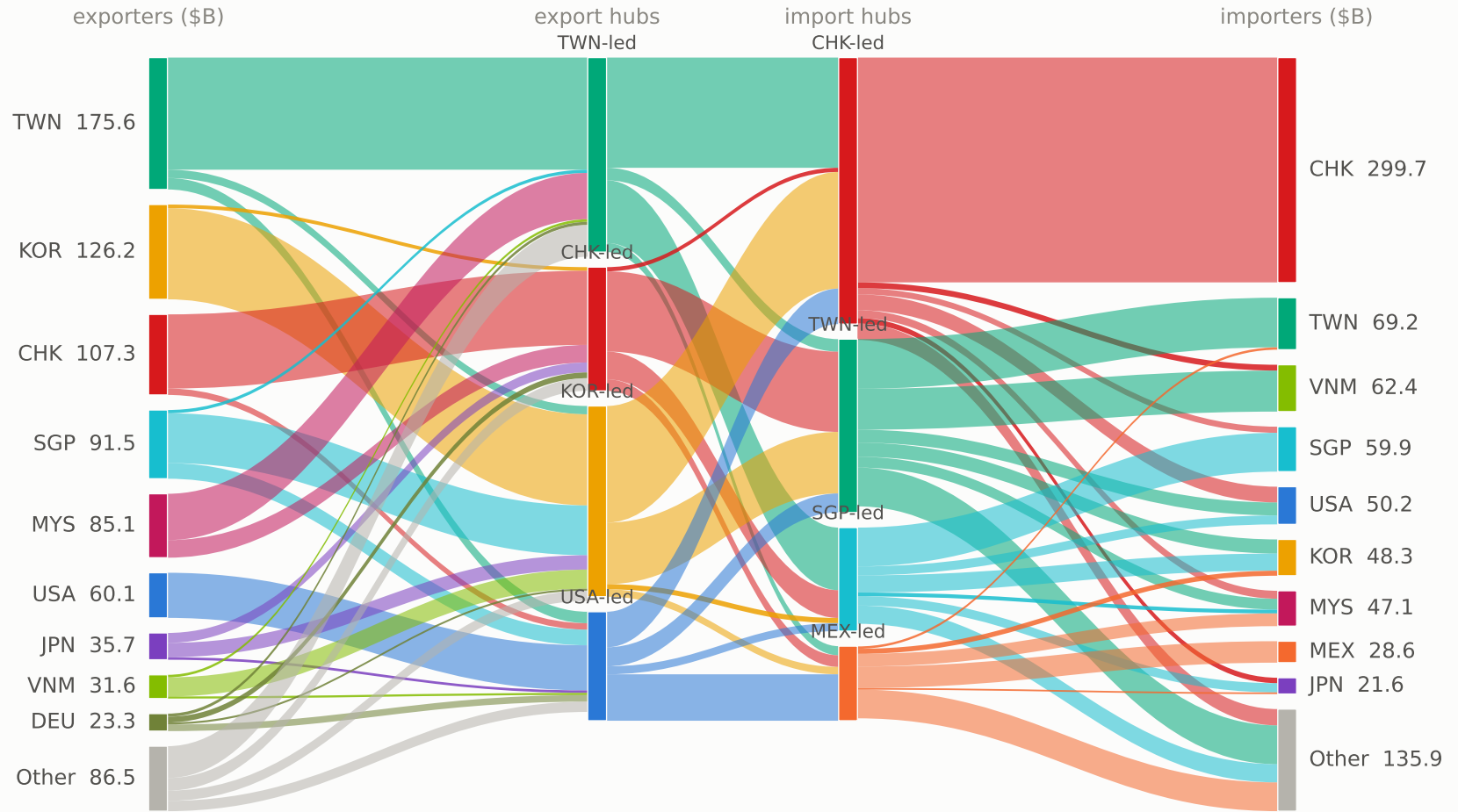


Stage 5 — Chips (logic, memory, discretives) (8542xx, 8541xx, media) — 2024
 through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 on Atlas bilateral data); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral, all countries.